Reg. No. | | | | | | | | | | | [12EC74]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER - 2015

ARM PROCESSORS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. What is meant by pipelining techniques? Explain the following pipelining hazards.

i. Read after-write pipelining hazards

ii. Pipelined branch hazard **10**

b. Explain briefly the instruction format of MUO [Manchester university zero] processor and also list the instructions supported by MUO processor. **04**

2. a. With a neat block diagram, explain the structure of the ARM cross development tool kit. **10**

b. Describe the principle features of the ARM architecture. **04**

3. a. Explain arithmetic and logical ARM instruction set with examples. **10**

b. Write an ARM assembly language program to perform. **04**

```
if ( i= =0)
{
        i= i+ 10;
}
```

4. a. With a neat block diagram explain 3-stage pipeline ARM organization with single cycle instruction. **10**

b. Explain in brief handshaking instruction between ARM and the co-processor. **04**

5. Explain the following thumb instruction formats.

i. Move shifted register. **05**

ii. Add/Subtract. **04**

iii. Load/ Store with immediate offset. **05**

6. a. With a suitable example, explain the most efficient way to use 'LOCAL VARIABLE TYPES' in the ARM. **10**

b. Write the differences between ARM and thumbs instruction set? **04**

LIBRARY

7. Explain the architecture of LPC17XX microcontroller with a neat block diagram and also write the properties and characteristics. **14**

8. a. Draw a neat diagram to interface a 4x4 matrix keypad to a LPC 17XX where keypad rows are connected to P0.23 to P0.26 keypad columns are connected to P0.19 to P0.22. **06**

b. Write an assembly /C language program to tern-ON and OFF a relay connected to P3.25 of LPC17XX and also draw its interfacing diagram. **08**